

**SRM Institute of Science and Technology
College of Engineering and Technology**

SCHOOL OF COMPUTING

MINI PROJECT REPORT

ODD Semester: 2024-2025

Lab code & Sub Name : 21CSS201T & Computer Organization and Architecture

Year & Semester : II & III

Project Title : Intruder Alarm System

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Particulars	Max. Marks	Marks Obtained		
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Review 1 and 2	05			
Demo verification & viva	03			
Project Report	02			
Total	10			

Date : 29/10/2024

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Signature :

INTRUDER ALARM SYSTEM

OBJECTIVE:

The objective of an intruder alarm system is to detect unauthorized access and promptly alert security personnel to prevent theft or harm.

ABSTRACT:

An Intruder Alert System is a security system designed to detect unauthorized entry or suspicious activity within protected areas such as a home, office, or restricted zone. It is designed to enhance security by detecting unauthorized access to premises. It uses sensors, such as motion detectors, infrared, or cameras, to monitor entry points or areas.

Features like facial recognition, biometric access control, and geofencing enhance security, while cloud-based platforms enable remote monitoring and control via smartphones or web interfaces. This technology is essential for safeguarding homes, offices, or high-security zones from potential threats. Some systems are equipped with smart features like facial recognition or voice commands, adding an extra layer of convenience and security.

Upon detecting an intrusion, the system immediately triggers audible alarms, sends real-time notifications to security personnel or homeowners and can even activate additional security measures, such as locking doors or contacting law enforcement. This project focuses on developing a cost-effective, real-time monitoring system to ensure reliable and responsive intrusion detection for homes or businesses.

INTRODUCTION:

- An intruder alarm system is a security system designed to detect unauthorized entry into a building or area. These systems use a combination of sensors, such as motion detectors, door/window contacts, and glass break sensors, to monitor entry points and specific areas.
- The primary function of the system is to safeguard property, assets, and people by alerting the owner or security personnel when an intrusion occurs.

- These systems are commonly used in residential, commercial, industrial, and government settings, providing real-time notifications of potential threats.

HARDWARE/SOFTWARE REQUIREMENTS:

- PIR Sensor
- 9V Battery
- BC547 Transistor
- Diode
- Switch
- 1K Ω Resistor
- Buzzer
- Connecting wires

CONCEPTS/WORKING PRINCIPLE:

The working principle of an intruder alarm system is based on detecting unauthorized entry through various sensors and triggering an alert mechanism. The system typically operates using the following key concepts:

1. Sensing Mechanism:

- **Motion Sensors:** Detect movement using technologies like infrared (PIR sensors), ultrasonic waves, or microwave signals. When movement is detected, the sensor sends a signal to the control panel.

2. Control Panel:

- The control panel is the system's brain, processing signals from the sensors and determining whether a breach has occurred. It also decides when to trigger alarms based on preset conditions.
- It may allow users to arm or disarm the system and can communicate with remote monitoring stations or mobile devices.

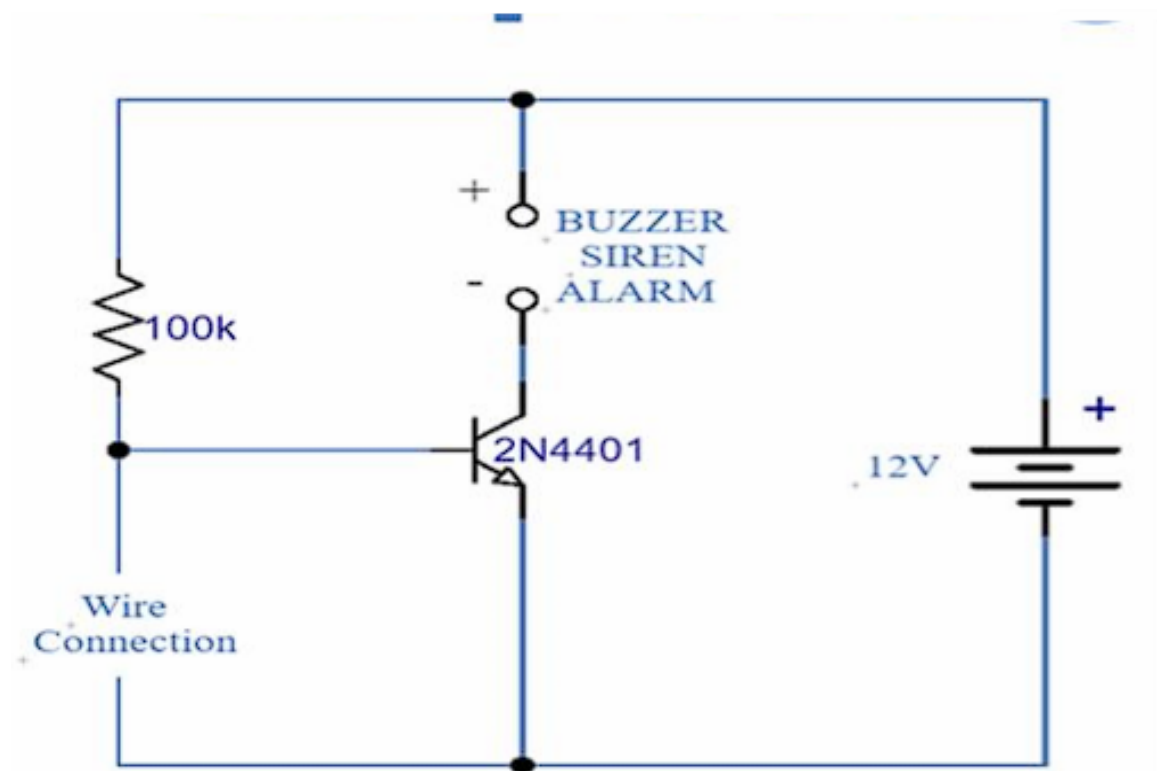
3. Alarm Activation:

- If the sensors detect an intrusion, the control panel activates the alarm, which can be in the form of audible sirens which are loud alarms that alert people nearby and scare off intruders.

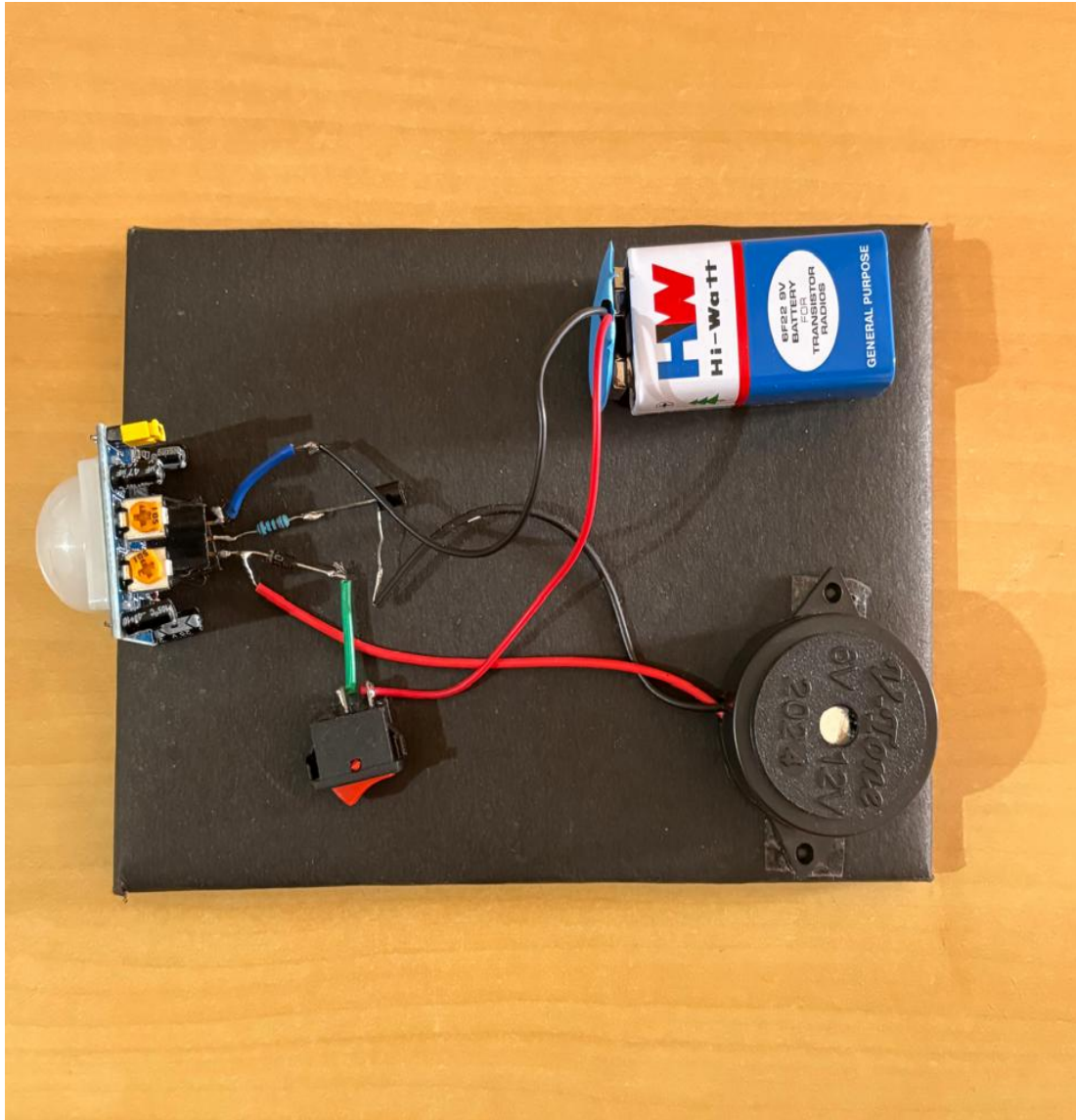
4. Power Supply:

- These systems usually rely on a mains power supply but are equipped with batteries to ensure continuous operation.

METHODOLOGY:



OUTPUT:



CONCLUSIONS:

In conclusion, an intruder alarm system plays a critical role in enhancing security by detecting unauthorized access, providing timely alerts, and deterring potential threats. Its effectiveness lies in its ability to protect property, ensure personal safety, and facilitate quick responses to security breaches, making it an essential component of modern security infrastructure.

REFERENCES:

- **A Smart Intruder Alert System Based on Microprocessor and Motion Detection-** <https://ieeexplore.ieee.org/document/8644109>
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- **Journal Articles:** J. P. Smith and R. G. Brown, "A study of wireless intruder detection systems," IEEE Transactions on Wireless Communications, vol. 12, no. 5, pp. 1025-1033, May 2017.